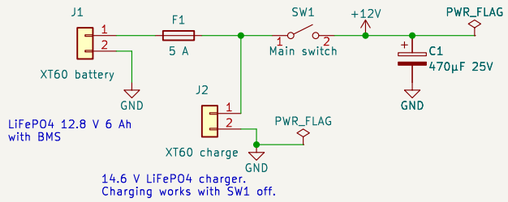
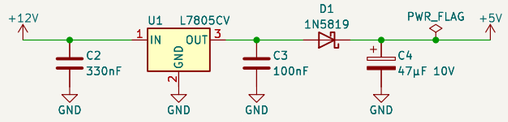
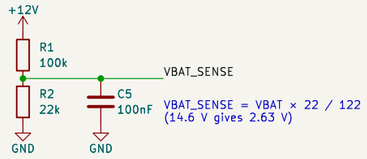
Battery, fuse, main switch



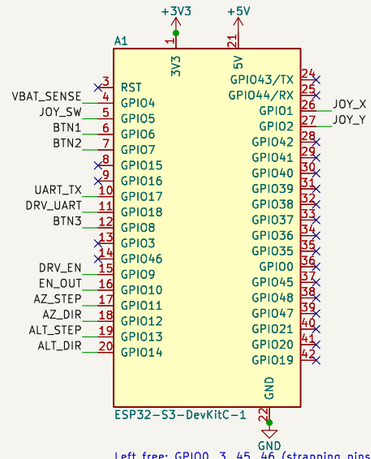
5 V supply for the ESP32 board



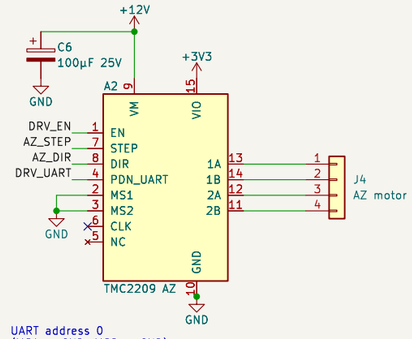
Battery voltage

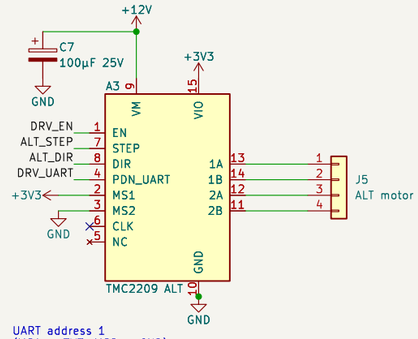


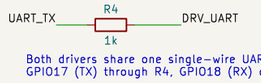
Controller



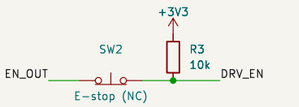
Stepper drivers

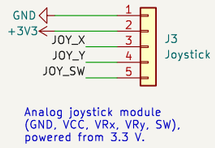


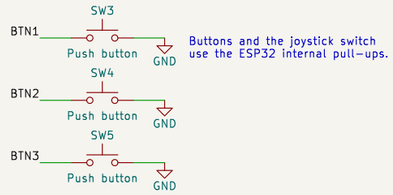




Manual controls







Notes

1. Battery, controller and drivers are all on the L: no cable crosses the azimuth joint.
2. +12V is the switched battery rail (12.0–14.6 V). It feeds the drivers and U1.
3. TMC2209 modules: VIO at 3.3 V (ESP32 logic level); motor current is set over UART.
   The header pin order differs between makers: wire by pin name.
4. Motor connectors: pins 1–2 = one coil, pins 3–4 = the other coil.
5. A1 pin numbers: 1–22 = header J1, 23–44 = header J3 of the DevKitC–1.
6. SW2–SW5 and J3 can sit on the case or on a short-cable hand unit.

All electronics on the L of the mount
ESP32-S3 controller, two TMC2209 drivers, 12.8 V LiFePO4 battery

Sheet: /
File: star-track.kicad_sch

Title: Star-Track mount electronics

Size: A3	Date: 2026-10-25	**Rev: 1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1